A Project Report on

Inclusive Career Path for Differently-Abled People

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "Inclusive Career Path for Differently-Abled People" is a bonafide work of **Harsh Raulgaonkar (23106082), Yash Sawant (23106009), Parth Rane (23106018), Sohan Vernekar (23106012)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

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Abstract

The increasing gap in employment accessibility for differently abled individuals remains a critical global challenge, driven by barriers in education, skill evaluation, and career guidance systems. Despite the presence of government initiatives in India, many individuals face difficulties due to inaccessible assessment methods and the absence of personalized career mapping. This project presents the design and development of an Inclusive Career Path platform, an accessibility-first web application aimed at empowering persons with disabilities through hands-free interaction and intelligent career recommendations. The system incorporates real-time gesture-based response capture, structured disability identification using weighted scoring, and personalized career domain extraction. Additionally, it provides tailored government scheme recommendations to enhance user support. Designed in compliance with WCAG 2.1 standards, the platform ensures inclusivity through features such as keyboard navigation, screen reader compatibility, high-contrast interfaces, and ARIA-based labeling. By enabling independent participation in career assessment, the platform promotes inclusive and equitable quality education (SDG 4), enhances access to decent work opportunities (SDG 8), and contributes to reducing inequalities (SDG 10). Simulated deployment results indicate improved decision-making efficiency, reduced dependency on external assistance, and increased engagement among users. This study introduces a scalable, technology-driven solution that redefines traditional career counselling into an inclusive, accessible, and empowerment-oriented ecosystem.

Keywords: Inclusive Careers, Accessibility Design, Gesture Interaction, Career Mapping.

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ABBREVIATION

AI	*Artificial Intelligence*
ML	*Machine Learning*
WCAG	*Web Content Accessibility Guidelines*
RPwD	*Rights of Persons with Disabilities*
HCI	*Human-Computer Interaction*
API	*Application Programming Interface*
HTML	*HyperText Markup Language*
CSS	*Cascading Style Sheets*
JS	*JavaScript*
IDE	*Integrated Development Environment*
UI	*User Interface*
UX	*User Experience*
SSD	*Solid State Drive*
CPU	*Central Processing Unit*
RAM	*Random Access Memory*
OS	*Operating System*
CV	*Computer Vision*
SDG	*Sustainable Development Goals*

Chapter 1
Introduction

1. Introduction

Employment accessibility remains a significant challenge for differently abled individuals, particularly in developing countries where structural, technological, and social barriers limit equal participation in education and the workforce. Although various government initiatives and welfare schemes exist to support persons with disabilities, access to accurate career guidance, structured assessments, and inclusive digital systems continues to be limited.

1.1. Background

In India, the Rights of Persons with Disabilities (RPwD) Act, 2016 recognizes 21 categories of disabilities and emphasizes equal opportunity in education and employment. However, practical implementation often faces gaps in accessibility, digital inclusion, and structured career mapping mechanisms. Individuals with visual, hearing, locomotor, neurological, or cognitive disabilities frequently encounter inaccessible interfaces, lack of assistive features, and the absence of personalized guidance systems tailored to their functional capabilities. Traditional career counselling systems are often generalized and do not incorporate adaptive assessment models that dynamically adjust based on disability profiles. Furthermore,

many online platforms lack compliance with accessibility standards, making them difficult to navigate using assistive technologies. This results in dependency on external assistance, lower participation rates in assessments, and limited awareness of suitable career pathways, government schemes, and support programs.

With advancements in web technologies, machine learning, and browser-based assistive capabilities, it is now possible to design systems that promote inclusive and independent interaction. Gesture recognition, real-time feedback mechanisms, and integrated audio guidance can eliminate barriers associated with keyboard or mouse dependency. By embedding accessibility into the core design rather than treating it as an additional feature, digital platforms can transform from passive information portals into empowerment tools.

The Inclusive Career Path for Differently Abled People Project introduces an accessibility-first web application designed to provide hands-free, guided, and structured career assessment. The system enables users to navigate and respond using gesture-based interaction supported by real-time recognition, along with continuous audio assistance to ensure clarity and independence. It follows a multi-stage evaluation process that collects user background information, conducts universal disability screening, performs targeted disability-specific assessments, and generates personalized career recommendations.

The platform emphasizes independence, structured evaluation, and secure result management. By integrating gesture-based input, automated question flow, and accessibility compliance features aligned with WCAG 2.1 standards, the system aims to remove barriers in digital career counselling. Through this approach, the project demonstrates how inclusive technology can bridge the gap between differently abled individuals and meaningful employment opportunities, fostering self-reliance and equitable workforce participation.

1.2. Objectives and Scope of the Project

The primary objective of this project is to develop an inclusive career guidance platform that enables differently abled individuals to interact using gesture-based inputs. The system aims to provide accurate assessment of user capabilities and deliver personalized career recommendations based on their abilities. It focuses on improving accessibility, reducing dependency on traditional input methods, and ensuring user-friendly interaction. Additionally, the project seeks to create a scalable, reliable, and efficient solution that enhances independence and decision-making for users.

The scope of this project includes designing and implementing a web-based platform that integrates gesture recognition technology with career assessment modules. It focuses on supporting users with physical disabilities by enabling hands-free interaction and providing accessible career guidance. The system can be extended to include machine learning models for improved recommendation accuracy, multilingual support, and integration with external job platforms. Future enhancements may also incorporate voice-based interaction and adaptive interfaces to further improve usability and accessibility.

1.3. Relevance to Sustainable Development Goals

The proposed Inclusive Career Path platform aligns closely with multiple United Nations Sustainable Development Goals (SDGs) by addressing critical challenges in accessibility and inclusion. The system contributes to **SDG 4 (Quality Education)** by enabling accessible, self-paced career assessment for differently abled individuals through gesture-based interaction and structured evaluation. It supports **SDG 8 (Decent Work and Economic Growth)** by providing personalized career recommendations and awareness of relevant government schemes, thereby improving employability and workforce participation. Furthermore, the platform promotes **SDG 10 (Reduced Inequalities)** by minimizing digital and structural barriers that restrict equal access to career guidance systems. By integrating accessibility standards such as WCAG 2.1 and enabling independent system interaction, the solution fosters inclusivity and equal opportunity. Overall, the project demonstrates how technology-driven, accessibility-first design can contribute to building a more equitable, inclusive, and sustainable digital ecosystem.

Chapter 2

Literature Survey/ Existing system

2. Literature Survey/ Existing system

The evolution of career guidance systems for differently abled individuals has progressed significantly over the past decades, driven by increasing awareness of accessibility, equal opportunity, and inclusive technology. Early approaches to career counselling were largely generalized and lacked consideration for disability-specific needs, resulting in limited participation from persons with disabilities. With advancements in assistive technologies, human-computer interaction, and web accessibility standards, modern systems are gradually shifting towards inclusive, adaptive, and technology-driven career assessment platforms.

2.1. History of Existing Systems

Early Steps (Pre-2010s):

In the initial phase, career guidance for differently abled individuals was primarily conducted through manual and facilitator-driven methods. Educational institutions, rehabilitation centers, and NGOs played a major role in providing counselling through interviews and paper-based assessments.

While these methods offered personalized attention, they were:

- Time-consuming and resource-dependent
- Limited in scalability
- Highly dependent on trained professionals

Digital systems during this period were minimal and mostly consisted of static websites or basic databases, which lacked accessibility features such as screen reader compatibility or alternative input methods. As a result, individuals with visual, locomotor, or cognitive impairments faced significant barriers in accessing these services.

Transition to Digital Platforms (2010s):

The 2010s marked a significant transition toward web-based career assessment systems. Educational platforms and online aptitude testing tools began to emerge, leveraging technologies such as HTML, PHP, and database systems for centralized data management.

These platforms introduced:

- Online career quizzes and aptitude tests
- Basic recommendation systems
- Centralized data storage

However, most systems were still:

- Generalized and non-adaptive
- Focused on academic performance rather than functional abilities
- Lacking accessibility compliance

Although this phase improved reach and digitization, it failed to adequately address disability-specific career mapping. Users still relied heavily on traditional input devices like keyboards and mice, which limited independent participation for differently abled individuals.

Modern Era (2020s–Present):

Recent advancements in assistive technology, machine learning, and web accessibility standards have transformed the development of inclusive digital platforms. Modern systems increasingly emphasize accessibility-first design, aligning with global standards such as WCAG 2.1.

Key technological developments include gesture recognition and computer vision systems for hands-free interaction, text-to-speech and audio guidance for visually impaired users, adaptive assessment frameworks based on user capabilities, browser-based applications eliminating dependency on complex installations.

These innovations enable users to interact independently without relying on traditional input methods. Platforms now incorporate real-time interaction, structured workflows, and personalized recommendations, improving usability and engagement.

2.2. Related Research

Evidence on disability inclusion highlights a significant gap in employment opportunities for persons with disabilities. Differently abled individuals face barriers in education, skill development, and workforce participation due to inaccessible systems and lack of structured career guidance.

Manual career counselling methods introduce dependency, delays, and reduced privacy, limiting independent participation. Research indicates that technology-driven platforms with accessibility features can improve user engagement, independence, and career mapping efficiency.

Modern studies emphasize the importance of intuitive interfaces, assistive interaction methods, real-time feedback, and secure data handling. Technologies such as gesture recognition, speech synthesis, and browser-based frameworks play a key role in enabling inclusive digital systems.

2.3. Reference Papers

Table 1 Comparative Analysis of Literature Survey Papers

Sr. No.	Paper	System	Features
1	T. Ahmetovic et al., "Investigating Gesture-Based Interaction for Accessible User Interfaces"	Gesture-based user interface system enabling accessible interaction without traditional input devices.	Hands-free interaction, real-time gesture detection, improved accessibility for motor impairments.
2	S. K. Kane et al., "Usable Gestures for Blind People"	Assistive gesture interaction system for visually impaired users.	Non-visual navigation, user-centered design, improved usability.
3	W3C, "Web Content Accessibility Guidelines (WCAG) 2.1"	Standard framework for accessible web design.	Screen reader support, keyboard navigation, inclusive UI design.

4	Google LLC, “MediaPipe Hands”	Real-time hand tracking system using computer vision.	21 key point detection, real-time processing, browser-based support.
5	J. Lazar et al., “Ensuring Digital Accessibility Through Process and Policy”	Accessibility-focused system design approach.	Inclusive design policies, usability improvements, compliance strategies.
6	S. Harper & Y. Yesilada, “Web Accessibility: A Foundation for Research”	Research framework for accessible web systems.	Adaptive interfaces, accessibility evaluation methods.
7	A. Sears & V. L. Hanson, “Representing Users in Accessibility Research”	User-centered accessibility system.	Focus on diverse user needs, inclusive interaction models.
8	G. W. Abowd et al., “Human-Computer Interaction for Assistive Technologies”	Assistive technology interaction systems.	Improved HCI models, enhanced usability for disabled users.

Chapter 3

Limitation of Existing system

3. Limitation of Existing system

3.1. Research Gap

Existing career guidance systems for differently abled individuals suffer from multiple limitations that restrict accessibility, usability, and effectiveness. These systems often fail to provide inclusive design, adaptive interaction, and structured assessment mechanisms, resulting in reduced participation and limited career mapping accuracy.

Despite advancements in digital platforms and assistive technologies, a significant gap exists in the development of fully inclusive and accessibility-first career guidance systems. Most existing solutions focus on general user groups and fail to address the specific needs of differently abled individuals. There is a lack of systems that integrate accessibility, personalization, and intelligent assessment into a single unified platform.

Current platforms do not effectively utilize modern technologies such as gesture recognition, real-time assistive interaction, and adaptive evaluation frameworks. Additionally, many systems lack compliance with accessibility standards, limiting their usability for individuals with visual, motor, and cognitive impairments.

Another major gap lies in the absence of structured, multi-stage assessment models that consider disability-specific capabilities. Existing systems often provide generalized recommendations, which may not align with individual abilities or real-world opportunities.

Furthermore, there is limited integration of government schemes and career resources, resulting in a lack of awareness among users.

The need for a comprehensive, scalable, and accessibility-first system is evident. Such a system should enable hands-free interaction, ensure WCAG compliance, provide personalized career mapping, and support secure and reliable assessment workflows. Addressing these gaps can significantly improve independence, participation, and employment opportunities for differently abled individuals.

3.2. Limitations Identified in Existing Systems

Table 2 Summary of Limitations in Existing Systems

Sr. No.	Category	Key Issues
1	Accessibility Limitations	Lack of accessibility-first design, poor assistive technology support
2	Input & Interaction Constraints	Dependence on keyboard/mouse, no hands-free interaction
3	Personalization Gaps	Generalized recommendations, no adaptive assessment
4	Accessibility Compliance	Non-adherence to WCAG, poor UI adaptability
5	Assistive Technology Integration	Limited use of gesture/speech technologies
6	Workflow Issues	Unstructured flow, risk of data loss
7	Awareness Gaps	Lack of government schemes and career info

Chapter 4

Problem Statement and Objective

4. Problem Statement

Contemporary career guidance systems face several limitations when addressing the needs of differently abled individuals.

4.1. List of Identified Key Problems

1. Lack of Accessibility-First Design:

Existing systems are not designed with accessibility as a primary focus, resulting in poor usability for individuals with visual, motor, and cognitive impairments.

2. Dependence on Traditional Methods:

Career counselling systems rely on manual processes, text-based interfaces, and facilitator-assisted evaluation, limiting scalability and user independence.

3. Fragmented User Experience:

The absence of integrated accessibility solutions leads to inconsistent and disconnected user experiences.

4. Generalized Career Recommendations:

Existing platforms provide recommendations based on general academic data rather than disability-specific capabilities, reducing effectiveness.

5. **Limited Assistive Interaction:**

Lack of gesture-based input and assistive technologies prevents users from interacting independently with the system.

6. **Absence of Adaptive Assessment:**

Current systems do not include structured or multi-stage evaluation frameworks, leading to incomplete or inaccurate assessment results.

7. **Lack of Workflow Structure:**

Poor system flow allows interruptions and incomplete assessments, affecting reliability.

8. **Accessibility Compliance Issues**

Many platforms do not follow accessibility standards, resulting in navigation and usability challenges.

4.2. Objectives

The objectives of the project are structured to address both functional and non-functional requirements:

1. **Design Accessibility-First Web Interface:**

Develop a web-based interface using HTML, CSS, and JavaScript that supports diverse users through high-contrast layouts, scalable typography, and intuitive navigation.

2. **Develop Modular System Architecture:**

Create a browser-based system that operates independently without complex backend dependencies, ensuring simplicity, privacy, and scalability.

3. **Implement Gesture-Based Interaction:**

Integrate real-time gesture recognition to enable hands-free system control and improve accessibility for users with motor impairments.

4. **Ensure Reliable Interaction Mechanism:**

Implement stable gesture detection and response validation to avoid unintended inputs and ensure accuracy.

5. **Develop Multi-Stage Assessment Framework:**

Design structured evaluation including profiling, disability screening, and adaptive questioning for accurate capability analysis.

6. **Enable Personalized Career Mapping:**

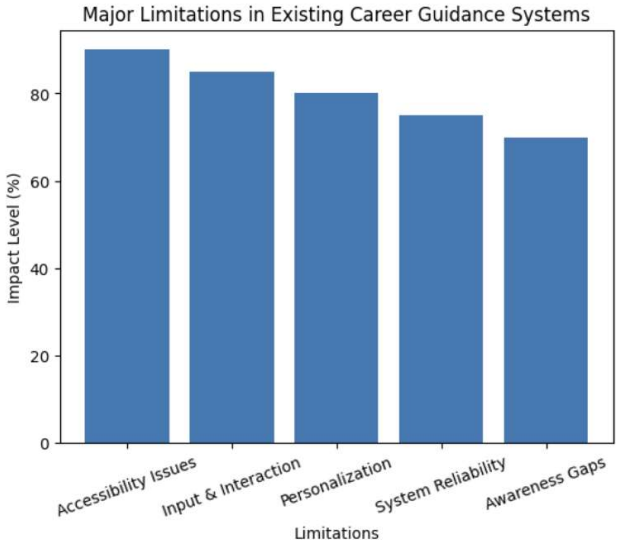
Create a system that generates career recommendations based on user abilities and assessment results.

7. **Ensure Transparency and Consistency:**

Implement deterministic scoring mechanisms to provide clear and explainable results.

8. Provide Accessible Output Presentation:

Present results in a clear, structured, and user-friendly format for better decision-making.



GRAPH 1 Major Limitations in Existing Career Guidance Systems

The graph illustrates the major limitations present in existing career guidance systems. Accessibility issues are the most significant barrier, followed by limitations in input and interaction methods. Lack of personalization and system reliability further reduce effectiveness, while awareness gaps highlight insufficient access to career-related information. Overall, the graph emphasizes the need for an accessibility-first, reliable, and adaptive system to improve user experience and enable independent career assessment for differently abled individuals.

Chapter 5

Proposed System

5. Proposed System

The proposed system constitutes a modular, browser-based web application designed to provide accessible and structured career guidance for differently abled individuals through an integrated, accessibility-first approach. The platform is engineered to support inclusive interaction by combining assistive technologies with a multi-stage assessment framework, enabling users to independently evaluate their capabilities and receive personalized career recommendations. The system primarily operates with a single-user interaction model, where individuals engage with the platform through guided assessment stages, eliminating dependency on external facilitators. The interface is designed to present clear, context-aware instructions and responses, ensuring usability across diverse disability categories.

5.1. Analysis / Framework / Algorithm

The system architecture is organized into multiple functional modules, including the User Interface Layer, Gesture Recognition Engine, Assessment Control Module, Scoring and Classification Engine, and Career Recommendation Module. Each module performs a specific role while maintaining seamless internal communication to ensure a structured workflow. The

User Interface provides an accessible environment with simplified navigation, high-contrast elements, and adaptive layouts. The Gesture Recognition Engine enables hands-free interaction by detecting predefined gestures such as Yes, No, and Idle, allowing users to navigate the system without traditional input devices.

The application is implemented using standard web technologies including HTML, CSS, and JavaScript, ensuring compatibility across modern browsers without requiring installation or specialized hardware. Real-time gesture recognition is achieved through browser-based computer vision frameworks, allowing the system to process webcam input locally. All computations, including assessment logic and gesture processing, are executed within the client-side environment, ensuring enhanced privacy and reduced system complexity. The modular design ensures scalability, maintainability, and consistent performance across different devices and user environments.

5.2. Design Details

The working flow diagram illustrates the operational sequence and interaction process within the proposed career guidance platform, highlighting the structured flow from user entry to generation of personalized career recommendations. The workflow is divided into multiple stages, ensuring a controlled and adaptive assessment process that supports independent user participation.

1. User Entry and System Initialization:

The process begins when the user accesses the platform through a web browser, where the system initializes required modules including the user interface, gesture recognition engine, and assessment control system. Users are introduced to the platform through clear instructions and are provided with guidance on interaction methods, including gesture-based input or alternative navigation options.

2. Pre-Assessment Profiling:

In this stage, the user provides basic information such as educational background, interests, and general preferences. This initial profiling helps in establishing a contextual understanding of the user and acts as a foundation for further assessment.

3. Universal Disability Screening:

Following profiling, the system initiates a universal screening phase designed to evaluate functional capabilities across multiple domains such as motor skills, communication ability, sensory perception, and cognitive processing. Questions are presented sequentially, and user responses are captured using predefined gestures, ensuring accessibility and ease of interaction.

4. Adaptive Disability-Specific Assessment:

Based on the results of the initial screening, the system dynamically activates relevant disability-specific assessment modules. This adaptive branching ensures that users are only presented with questions relevant to their condition, reducing assessment time and improving accuracy.

5. Real-Time Gesture-Based Interaction:

Throughout the assessment process, the system continuously monitors and processes user gestures through the webcam. Gesture inputs are validated using stability and confidence thresholds to ensure accurate response capture.

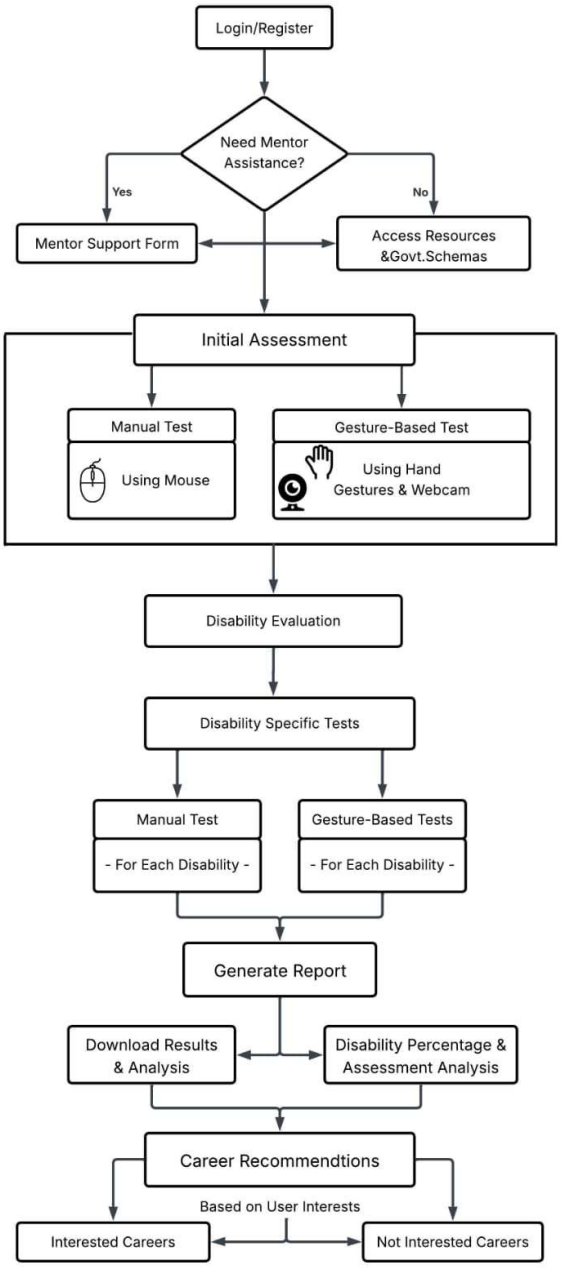


Figure 1 Flowchart

5.3. Methodology

1. **Scoring and Capability Analysis:**

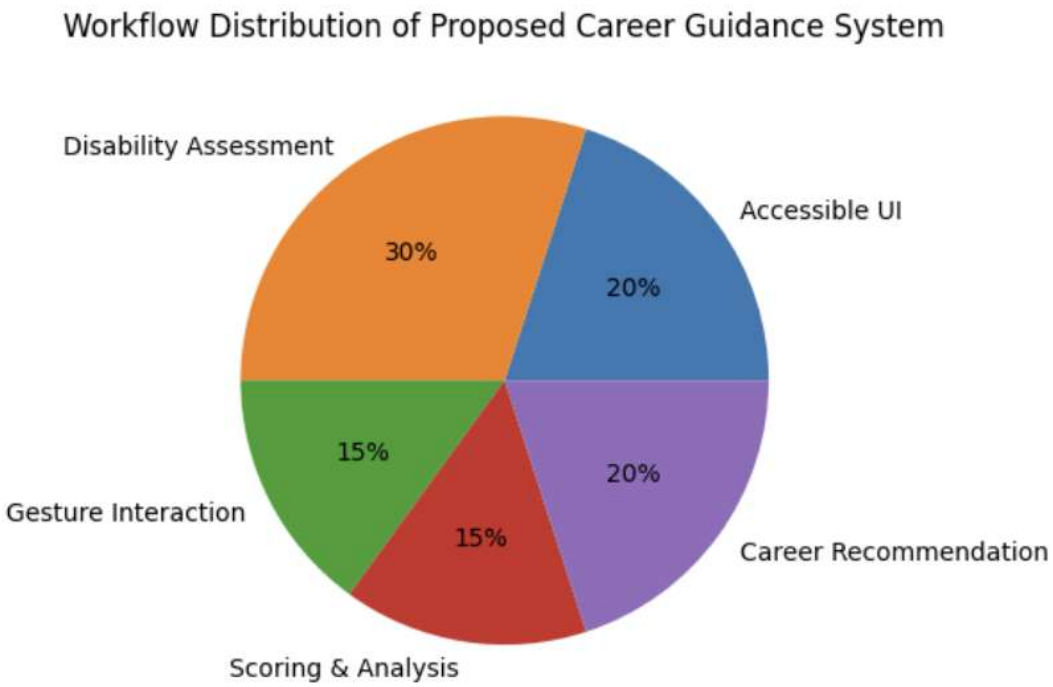
Once the assessment is completed, the system processes user responses through a rule-based scoring mechanism. Weighted values are assigned to different responses, allowing the system to identify dominant functional capabilities. This analysis forms the basis for mapping user strengths to appropriate career domains.

2. **Career Recommendation Generation:**

Based on the calculated scores, the system generates personalized career recommendations aligned with the user's capabilities. Recommendations are presented in a structured format, including primary career options, alternative paths, and relevant guidance. The output is designed to be clear, accessible, and easy to interpret.

3. **Result Display and Completion:**

The final stage involves displaying the assessment results on the interface, allowing users to review their recommended career paths. The system ensures that results are presented without ambiguity and provides a complete summary of the evaluation.



GRAPH 2 Distribution of Assessment Components

Chapter 6

Experimental Setup

6. Experimental Setup

The experimental platform comprises a class of mid-range personal computing devices selected to represent typical end-user environments for browser-based applications.

6.1. Software and Hardware Setup

The system requires processors equivalent to Intel Core i5 or AMD Ryzen 5 series, providing sufficient computational power for real-time gesture recognition and web-based processing. A minimum of 8 GB RAM is required to support concurrent browser operations, real-time video processing, and development tools such as code editors and debugging environments. Storage requirements include at least 256 GB of solid-state storage to ensure faster system performance and application loading times.

The system must support integrated or external webcams to enable real-time gesture recognition functionality. The operating system layer includes compatibility with Windows 10 or later, macOS, and Linux distributions such as Ubuntu, ensuring flexibility across development and user environments. Modern web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox are required, as the application relies on browser-based APIs for gesture detection and user interaction.

To implement the proposed accessibility-first architecture, the software stack integrates modern web technologies, browser-based frameworks, and version control systems to ensure efficient development and deployment.

Integrated Development Environment: Visual Studio Code (version 1.70 or later) serves as the primary development environment, offering support for JavaScript debugging, live server functionality, and extension-based enhancements for web development. Browser developer tools are utilized for inspecting DOM elements, debugging scripts, and monitoring performance.

Frontend Framework: The user interface is developed using HTML5, CSS3, and JavaScript (ES6+), providing a lightweight and responsive web-based interface. The design follows accessibility-first principles, incorporating semantic HTML and ARIA attributes to support assistive technologies.

Gesture Recognition Framework: The system integrates MediaPipe Hands or similar browser-based computer vision libraries to enable real-time hand tracking and gesture recognition. This framework processes webcam input directly within the browser without requiring external installations.

Version Control: Git is used for version control, with repositories managed through platforms such as GitHub. Branching strategies ensure structured development workflows, and version tracking supports maintainability and collaboration.

6.2. Details of Input to System

The system accepts user input in multiple forms to support accessibility and inclusive interaction. Primary input is captured through real-time gesture recognition using webcam-based hand tracking, where predefined gestures such as Yes, No, and Idle are used for navigation and response selection.

Additional input includes user-provided information during the pre-assessment stage, such as educational background, interests, and preferences. During the assessment phase, responses are captured through gesture-based interaction or alternative keyboard navigation where required.

The system processes this input through structured workflows, including universal disability screening and adaptive disability-specific evaluation. All input data is handled locally within the browser environment, ensuring privacy and minimizing dependency on external systems.

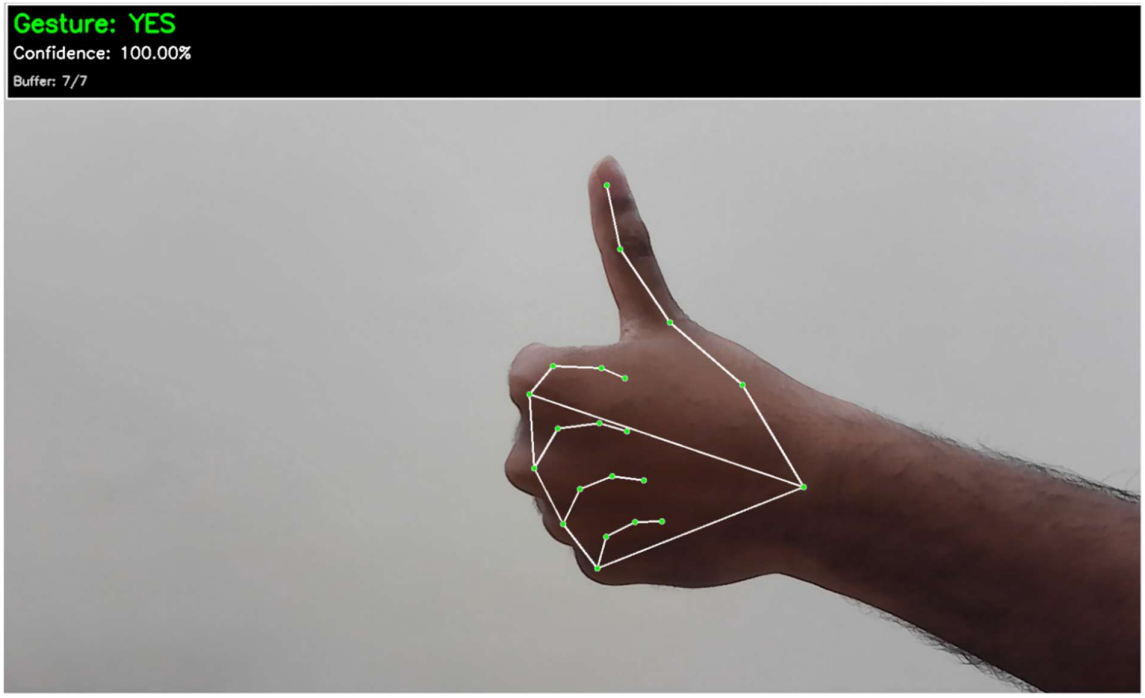


Figure 2 Gesture Recognition Output – “YES” Detection with High Confidence

This image demonstrates successful gesture recognition where the system detects a positive hand sign with high confidence accuracy.The model accurately tracks finger positions and interprets the gesture as affirmative, ensuring reliable user interaction and response.

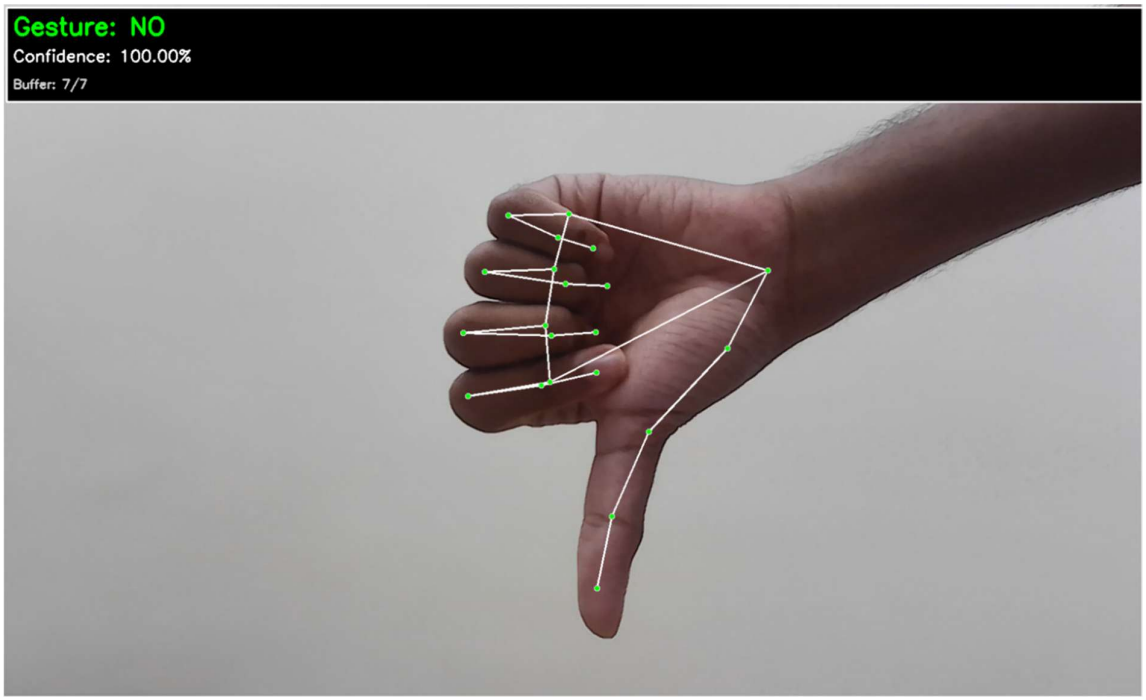


Figure 3 Gesture Recognition Output - "NO" Detection with High Confidence

The image on the previous page shows the system identifying a negative hand gesture using real-time tracking and classification with high confidence level. The detection algorithm effectively distinguishes finger orientation and interprets the gesture as rejection, enabling precise user feedback and control.

Figure 4 Assessment Result Showing Detected Disabilities and Functional Analysis

This image presents the assessment result interface displaying detected disabilities and functional analysis based on user interaction inputs clearly. The system generates structured outputs and recommendations, helping users understand their abilities and receive personalized career guidance effectively.

The user can then view personalized career recommendations based on the assessment results and responses provided. The user can further explore detailed information about each career by clicking on the suggested career options displayed on the interface.

Chapter 7

Results and Discussion

7. Results and Discussion

This chapter presents the outcomes of the Inclusive Career Path system and analyses its effectiveness across key functional stages, including user profiling, disability assessment, gesture-based interaction, and overall career recommendation generation. The results validate the system's ability to provide an accessible and structured platform for career guidance while enhancing independent participation for differently abled individuals.

7.1. Results

The developed system effectively bridges the gap between differently abled users and structured career guidance by offering an integrated and accessible digital platform for independent evaluation and decision-making. The integration of browser-based gesture recognition significantly improved user interaction by enabling hands-free navigation and reducing dependency on traditional input devices.

The implementation of real-time processing and local execution ensured reliable system performance and enhanced data privacy, as no external data transmission was required.

Testing across various scenarios demonstrated that the system provides a smooth and responsive user experience, with minimal latency in gesture detection and question transitions. The structured workflow ensured that users could not skip stages, resulting in complete and consistent assessments. The modular design further supported stable performance across different devices and environments.

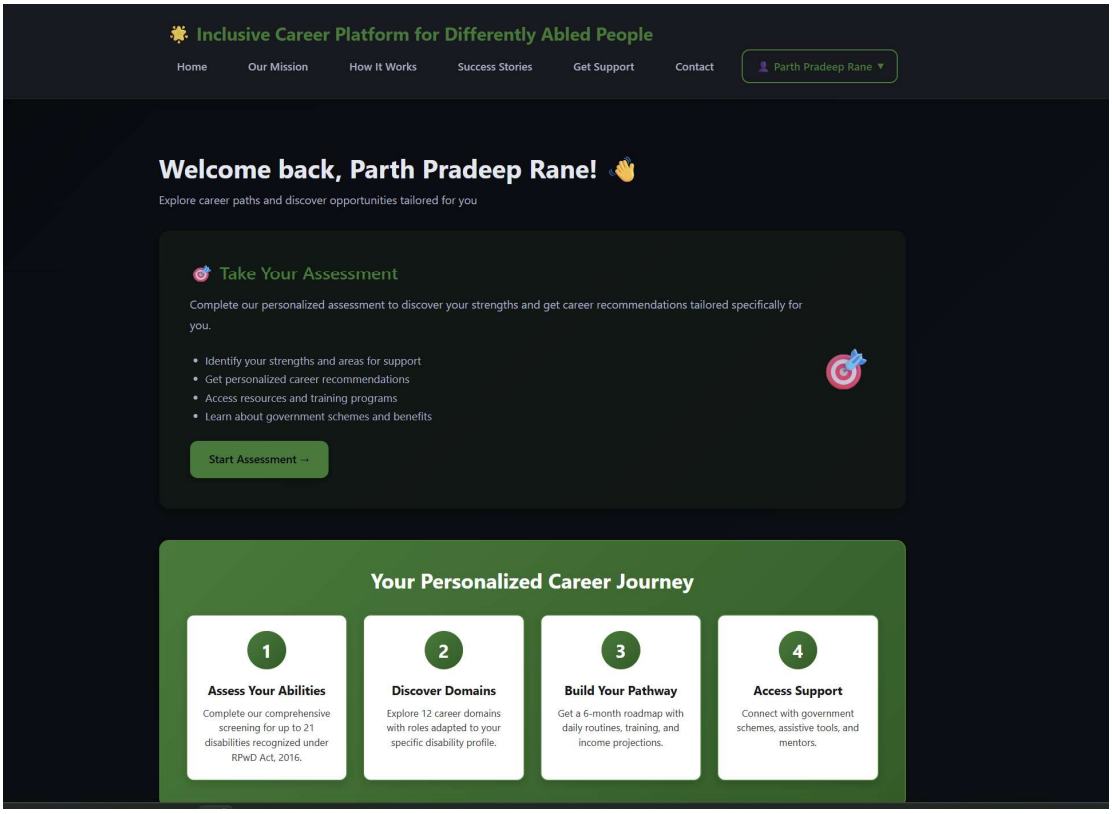


Figure 5 Dashboard Interface Showing User Interaction and Career Guidance Features

The dashboard provides accessible navigation to start assessments, explore resources, and view personalized career pathways, enhancing user engagement and usability.

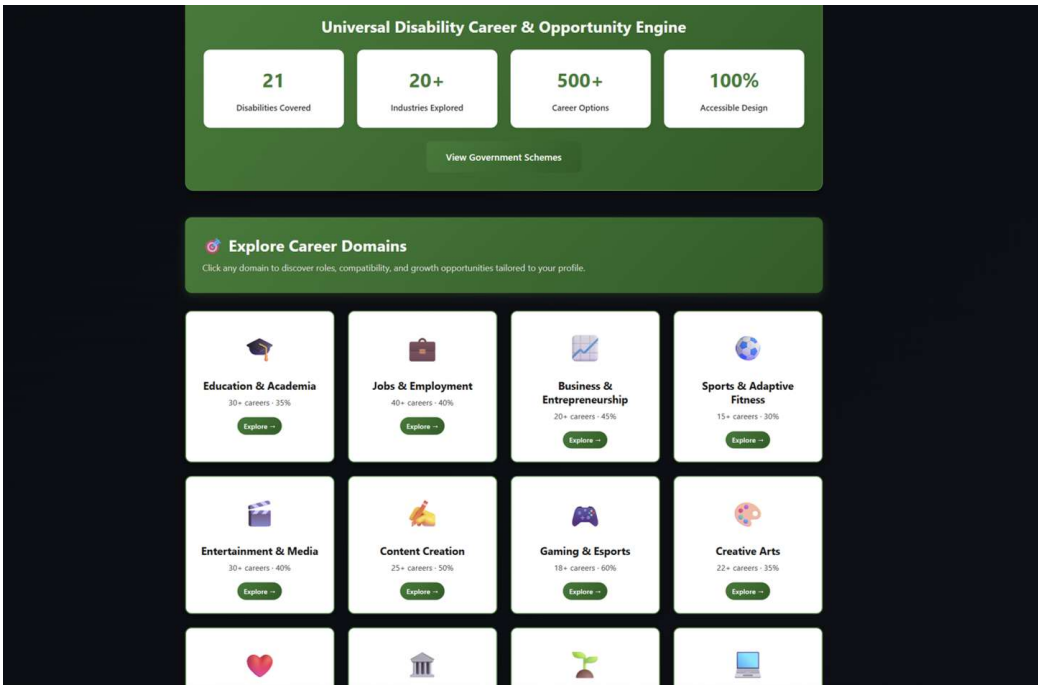


Figure 6 Dashboard Section Displaying Career Domains and Opportunity Exploration

This section of the dashboard enables users to explore diverse career domains with structured navigation, supporting informed decisions through accessible and tailored opportunities.

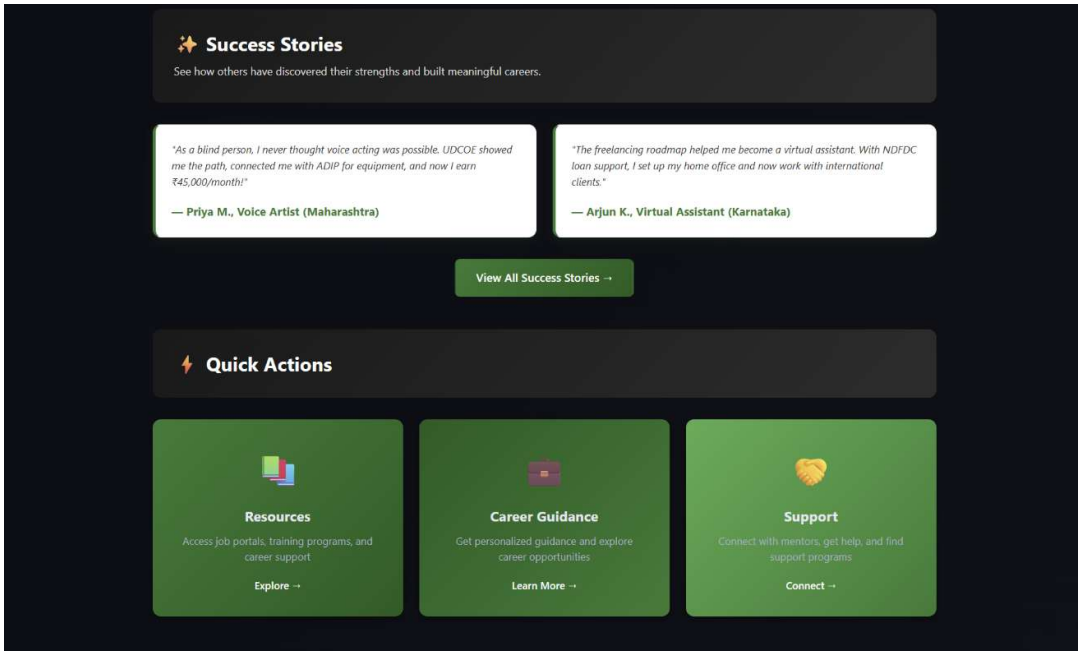


Figure 7 Interface Displaying Success Stories and Quick Access Panel

This section highlights success stories and provides quick access to resources, career guidance, and support, enhancing motivation and user engagement effectively.

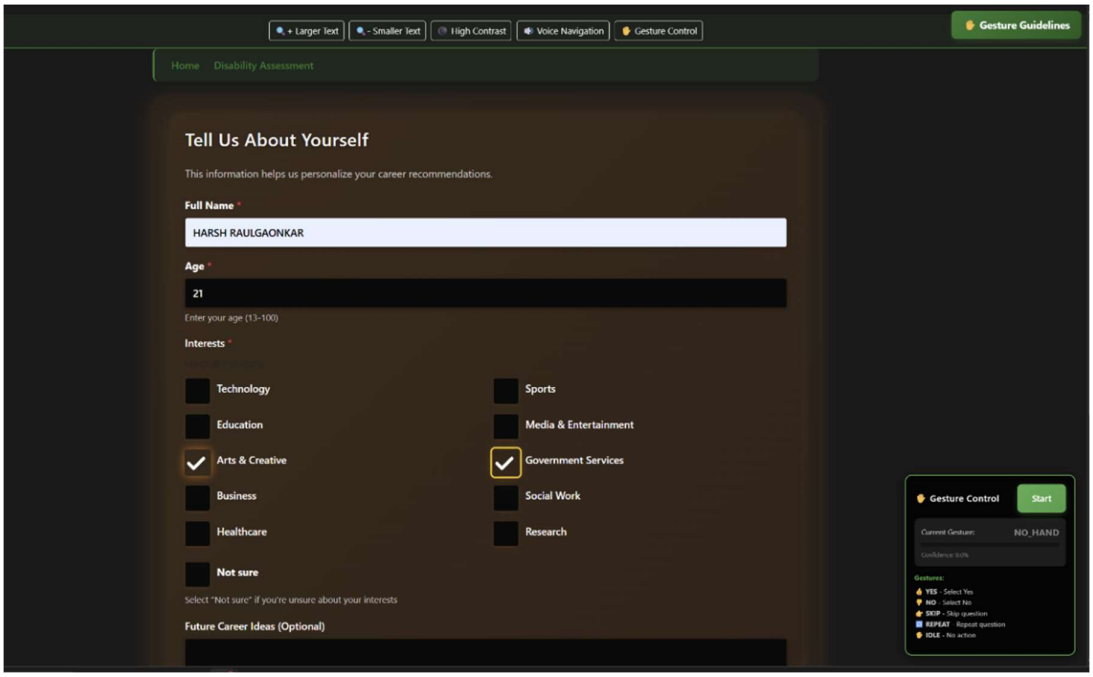


Figure 8 Pre-Assessment Interface Capturing User Details and Interests for Mapping

This interface captures user details and interests, enabling personalized career mapping through accessible inputs and structured data collection for accurate assessment.

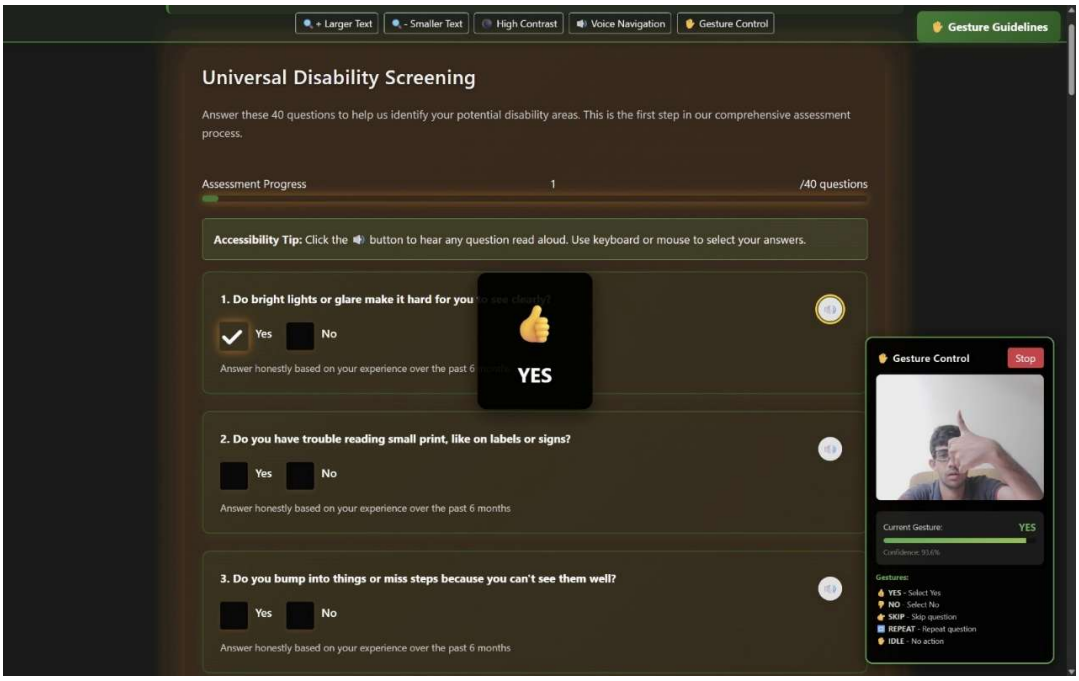


Figure 9 Universal Disability Assessment with Gesture-Based Real-Time Response

This interface conducts structured disability screening using gesture-based input, enabling real-time response detection and accessible interaction for independent user participation.

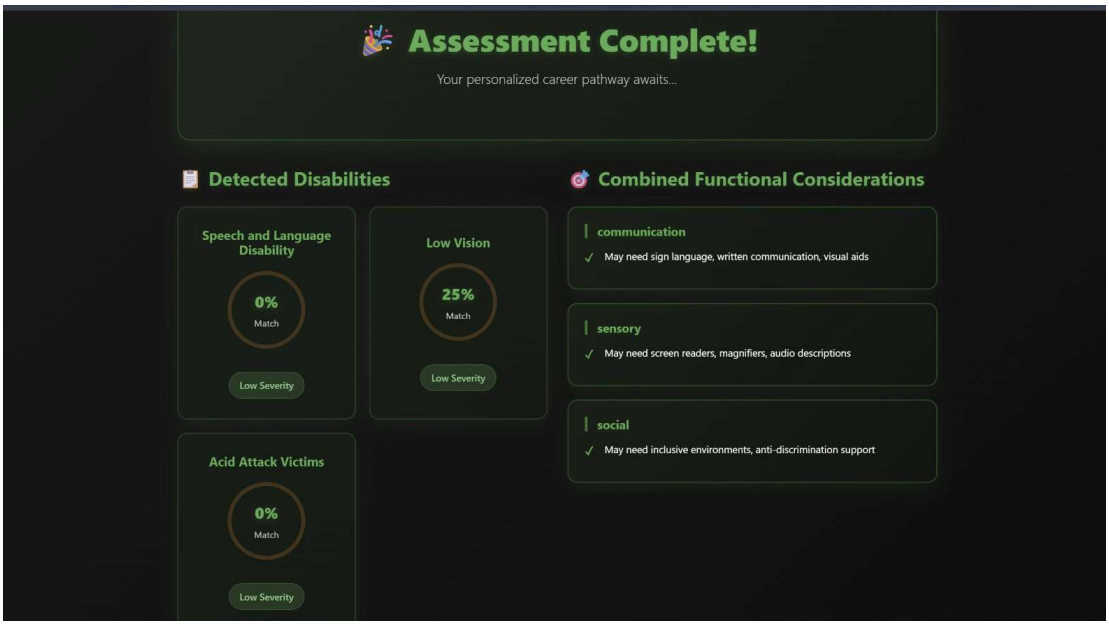


Figure 10 Assessment Results Displaying Detected Disabilities and Functional Analysis

This interface presents detected disabilities and functional insights, enabling users to understand assessment outcomes and support needs for informed career decisions.

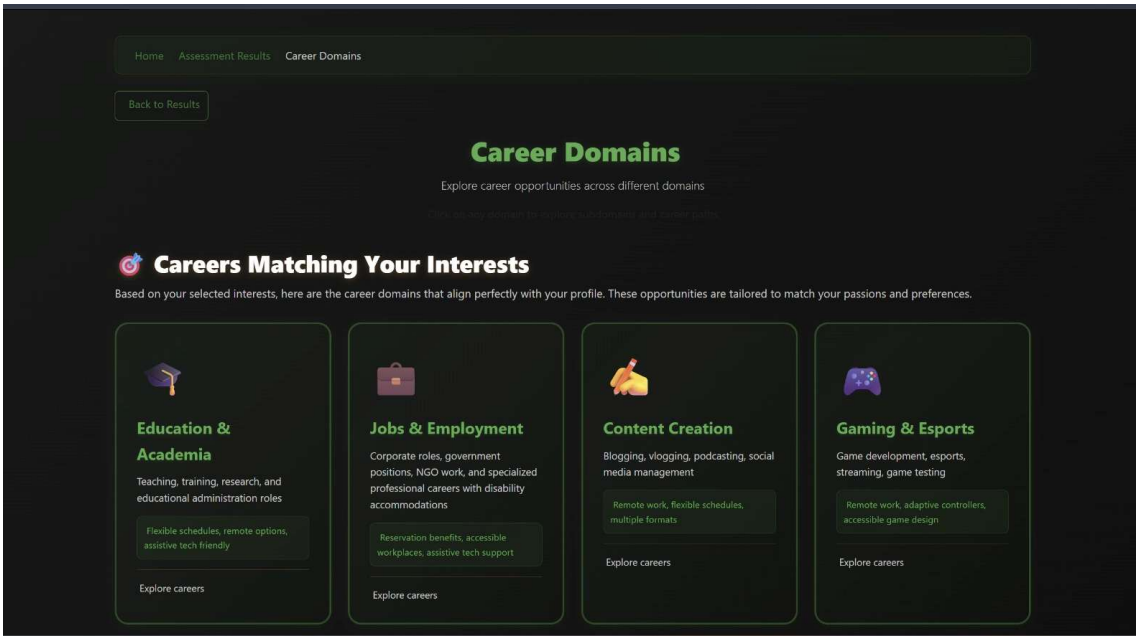


Figure 11 Career Domain Interface Displaying Personalized Career Recommendations Based on User Interests

This interface presents personalized career domains aligned with user interests, enabling exploration of suitable opportunities with accessible and structured guidance.

7.2. Discussion

The results of the "Inclusive Career Path for Differently-Abled People" system validate the effectiveness of integrating accessibility-first design, gesture-based interaction, and structured assessment mechanisms to enable independent career guidance for individuals with disabilities.

Key Performance Outcomes:

- **Improved Accessibility and Independent Interaction:**

Gesture-based interaction and accessibility-focused design enabled users to navigate without traditional input devices, reducing dependency and improving usability.

- **Structured and Adaptive Assessment Framework:**

The multi-stage evaluation process ensured accurate capability identification and provided personalized career recommendations.

- **Efficient and Real-Time Interaction:**

Browser-based gesture recognition enabled real-time response capture with minimal latency, ensuring smooth and stable interaction.

- **Enhanced User Experience and Usability:**

High-contrast visuals, simplified layouts, and guided navigation improved engagement and allowed users to complete assessments independently.

- **Privacy-Focused and Scalable Architecture:**

Local browser-based processing enhanced data privacy, while the modular design supports scalability and easy maintenance.

7.3. Limitations of Proposed System

- **Limited Real-World Validation:**
  The system is tested in controlled environments and may require further validation with diverse real-world user groups.
- **Restricted Career Dataset Coverage:**
  Career recommendations are based on predefined domains, which may not cover all emerging or niche career opportunities.
- **Lack of Continuous Learning Mechanism:**
  The system does not adapt dynamically over time based on user behavior or feedback.

- **Dependence on User Input Accuracy:**
  Assessment results rely on user responses, which may vary in accuracy and affect recommendation quality.
- **Limited Integration with External Systems:**
  The platform currently operates independently without real-time linkage to job portals, training platforms, or external databases.

Chapter 8

Conclusion and Future Work

8. Conclusion

The Inclusive Career Path system represents a significant advancement toward accessibility-driven digital transformation in the domain of career guidance for differently abled individuals. By integrating assistive technologies with structured assessment frameworks, the platform effectively addresses challenges such as lack of accessibility, generalized career recommendations, and dependency on traditional input methods. Through an accessibility-first design, the system enables independent participation, allowing users to interact seamlessly without reliance on external assistance.

8.1. Conclusion of the Project

Developed using modern web technologies, the application achieves efficient performance and broad compatibility across devices and platforms. The integration of real-time gesture recognition enables hands-free interaction, while the browser-based architecture ensures minimal deployment complexity and enhanced data privacy.

The system's modular design, comprising gesture recognition, assessment control, and career mapping components, ensures scalability, maintainability, and consistent user experience.

The platform follows a structured architectural approach that separates interface, interaction logic, and assessment processing, ensuring clarity in system design and ease of future enhancements. This approach enhances system reliability and supports accurate evaluation of user capabilities while maintaining secure handling of user data and controlled workflow execution.

In practical terms, the proposed system can significantly improve accessibility in career assessment processes by reducing dependency, increasing user engagement, and providing personalized guidance aligned with individual capabilities. Future advancements may incorporate machine learning for adaptive recommendation systems and multimodal interaction methods, demonstrating how technology can promote inclusivity and empower differently abled individuals in their professional development.

8.2. Future Enhancements

Future developments of the Inclusive Career Path system will focus on enhancing intelligence, interaction capabilities, and system scalability to further improve user experience and effectiveness. One major enhancement includes the integration of machine learning models to enable predictive career recommendations based on user behavior, historical data, and evolving job market trends. This will allow the system to provide more dynamic and personalized guidance.

Additionally, multimodal interaction capabilities such as voice recognition and text-to-speech systems will be incorporated to support users with diverse disabilities. These features will complement gesture-based interaction, creating a more flexible and inclusive user interface that adapts to individual needs and preferences.

To improve accessibility on a broader scale, multilingual support will be implemented, allowing users to interact with the system in regional languages. This will enhance usability and promote adoption across diverse user groups with varying linguistic backgrounds.

Further enhancements may include integration with external platforms such as job portals, vocational training systems, and government databases to provide real-time opportunities and resources. User profile management and progress tracking features can also be introduced to support long-term career development and repeated system usage.

References

[1] World Health Organization, World Report on Disability. Geneva, Switzerland: WHO Press, 2011.

[2] T. Ahmetovic, D. Gleason, C. Ruan, and J. O. Wobbrock, "Investigating Gesture-Based Interaction for Accessible User Interfaces," in Proc. ACM SIGACCESS Conf. Computers and Accessibility (ASSETS), 2020, pp. 1–12.

[3] S. K. Kane, J. O. Wobbrock, and R. E. Ladner, "Usable Gestures for Blind People: Understanding Preference and Performance," in Proc. SIGCHI Conf. Human Factors in Computing Systems, 2011, pp. 413–422.

[4] W3C, "Web Content Accessibility Guidelines (WCAG) 2.1," World Wide Web Consortium, 2018.

[5] Google LLC, "MediaPipe Hands: Real-Time Hand Tracking, "Google Research. [Online]. Available: https://developers.google.com/mediapipe

[6] J. Shotton et al., "Real-Time Human Pose Recognition in Parts from Single Depth Images,"

in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2011, pp. 1297–1304.

[7] R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed. Cham, Switzerland: Springer, 2022.

[8] N. Patel, S. Gupta, and R. Verma, "Location-Based Services in Mobile Applications: Architecture and Implementation Patterns,"

in Proc. IEEE Int. Conf. Mobile Computing, Bangalore, India, 2023, pp. 234–241.

[9] S. R. Klemmer, B. Hartmann, and L. Takayama, "How Bodies Matter: Five Themes for Interaction Design,"in Proc. ACM Conf. Designing Interactive Systems (DIS), 2006, pp. 140–149